

ADITYA CHOUBEY

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University of Edinburgh, Edinburgh, United Kingdom

PERSONAL STATEMENT

Computer science engineer specialising in backend and systems engineering, with a strong focus on distributed systems, concurrency, and performance. Experienced in C/C++ and building scalable, fault-tolerant systems, including distributed runtimes and file systems. Proven ability to design and debug complex systems, with an emphasis on correctness, efficiency, and clean abstractions. Interested in solving challenging infrastructure problems and building high-performance systems at scale.

EDUCATION

University of Edinburgh

Edinburgh, United Kingdom

MSc in Computer Science

September 2025 - September 2026

- **Dissertation Topic:** Optimising Network Packet Processing on Raspberry Pi (Debian) OS via Linux Kernel Bypass Techniques.
- **Modules:** Distributed Systems, Design and Analysis of Parallel Algorithms, Message-Passing Programming, Parallel Design Patterns, Parallel Programming Languages and Systems, Security Engineering, Applied Cloud Programming.

Manipal Institute of Technology, MAHE

Karnataka, India

BTech in Information Technology

September 2020 - August 2024

with Minor Specialisation in Computer Graphics

- **GPA:** 8.96 / 10
- **Modules:** Computer Network Protocols, Computer Organisation and Microprocessor Systems, Design and Analysis of Algorithms, Operating Systems, Embedded Systems, Data Structures, Object Oriented Programming, Software Engineering, Digital Image Processing, Neural Networks and Fuzzy Logic.

WORK EXPERIENCE

Traceable by Harness

Bengaluru, India

Software Engineer-1

January 2024 – July 2025

- Designed and operated scalable backend systems processing large volumes of API traffic in Linux-based production environments, with a focus on performance, reliability, and data quality.
- Developed heuristics to identify and block invalid APIs by analysing customer traffic patterns, redesigning the entity creation pipeline to improve correctness and efficiency.
- Achieved a 78% reduction in false APIs processed, 43% reduction in storage costs, and 21% improvement in response times through targeted system optimisations.
- Built a flexible, attribute-based filtering system exposed via GraphQL, enabling dynamic configuration without engineering intervention while maintaining low-latency performance.
- Extended traffic classification logic to accurately differentiate WebSocket APIs from HTTP/S traffic, improving observability and correctness for real-time applications.
- Collaborated closely with platform, product, and customer-facing teams to deploy and maintain reliable systems under production workloads.

Artificial Intelligence and Robotics Laboratory (AIRL), Aerospace Dept., IISc Bengaluru, India

Systems Design Intern

January 2023

- Developed embedded firmware for a real-time antenna tracking system, implementing closed-loop servo motor control to maintain antenna alignment within 2 degrees during dynamic UAV manoeuvres.
- Integrated firmware with a Python-based operator interface, enabling closed-loop control, real-time system monitoring, and manual override during live flight tests.
- Designed and deployed a low-latency video streaming pipeline with end-to-end latency under 10 ms, supporting reliable long-range UAV surveillance and improved situational awareness.

KEY ACHIEVEMENTS

Parallel Actor Framework for Predator–Prey Simulation

February 2026 - March 2026

Distributed actor-based runtime in C++ for scalable, message-driven computation across multi-node systems.

- Designed and implemented a distributed actor runtime in C++ using MPI to enable asynchronous, message-driven execution across multiple nodes, addressing the need for scalable coordination in compute-heavy simulations.
- Architected a partitioned 2D computation model with locality-aware actor placement, reducing inter-process communication overhead and improving scalability under increasing workloads.
- Developed robust message routing and actor lifecycle management, resolving issues such as ID collisions and ensuring correctness in a distributed environment.
- Diagnosed and fixed MPI memory leaks and communication bottlenecks through systematic debugging and profiling, improving system stability during long-running executions.
- Parallelised a Lotka–Volterra workload to validate the framework, demonstrating effective scaling and correctness across distributed processes.

FactorZero

November 2025

Fault-tolerant distributed file system and parallel workload engine

- Designed and implemented a distributed file system with whole-file caching, primary–backup replication, and heartbeat-based failover to ensure reliability under node failures.
- Built a coordinator–worker execution framework for large-scale primality testing, enabling parallel task distribution with fault recovery.
- Implemented idempotent gRPC retry semantics and Lamport clocks for consistent, ordered result aggregation in unreliable network conditions.
- Verified system correctness using Chandy–Lamport snapshots and conducted scalability benchmarking to optimise throughput under varying failure scenarios.

Sandpile Simulation via 2D Decomposition using MPI

November 2025

Distributed simulation system using MPI on large-scale HPC infrastructure

- Re-architected a sequential 2D simulation into a distributed-memory system using domain decomposition and MPI, enabling scalable execution across hundreds of processes.
- Implemented halo exchange for inter-process communication and ensured correctness via validation against serial execution.
- Achieved strong scaling up to 1024 processes on ARCHER2, significantly reducing runtime while identifying communication overhead as the primary bottleneck.

Project MANAS, MAHE

May 2021 - May 2023

MIT's official AI and Robotics team developing autonomous ground and aerial systems.

- Developed the control stack for the self-driving car and other autonomous robots in ROS, focusing on reliable navigation and system integration.
- Designed the imaging pipeline and ODCL web application for the AUVSI SUAS 2022 competition, enabling manual classification of UAV image streams and automated submission of processed data to the interoperability server. Also contributed to the overall flight stabilisation system of the UAV.
- Achieved strong results at AUVSI SUAS 2022, ranking 18th of 71 teams globally, including 2nd in Flight Readiness Review and 10th in Technical Design Review.

TECHNICAL SKILLS & CERTIFICATIONS

- **Programming & Scripting:** C, C++, Python, Java, Bash.
- **Systems & Low-Level:** Linux, POSIX systems, processes & memory management, system-level debugging
- **Distributed Systems & Parallel Computing:** MPI, gRPC, fault tolerance, replication, consistency models, performance analysis
- **Cloud & Infrastructure:** AWS (EC2, S3, VPC), Google Cloud Platform, Microsoft Azure
- **Tools & DevOps:** Git, Docker, Kubernetes, CI/CD pipelines
- **Certifications:** Linux Kernel Development (LFD103), Microsoft .NET Full-Stack Developer Certification.
- **Extracurricular:** Founding member of Adventuring Guild Mumbai. Built and scaled a non-profit community, driving collaboration, event organisation, and stakeholder coordination